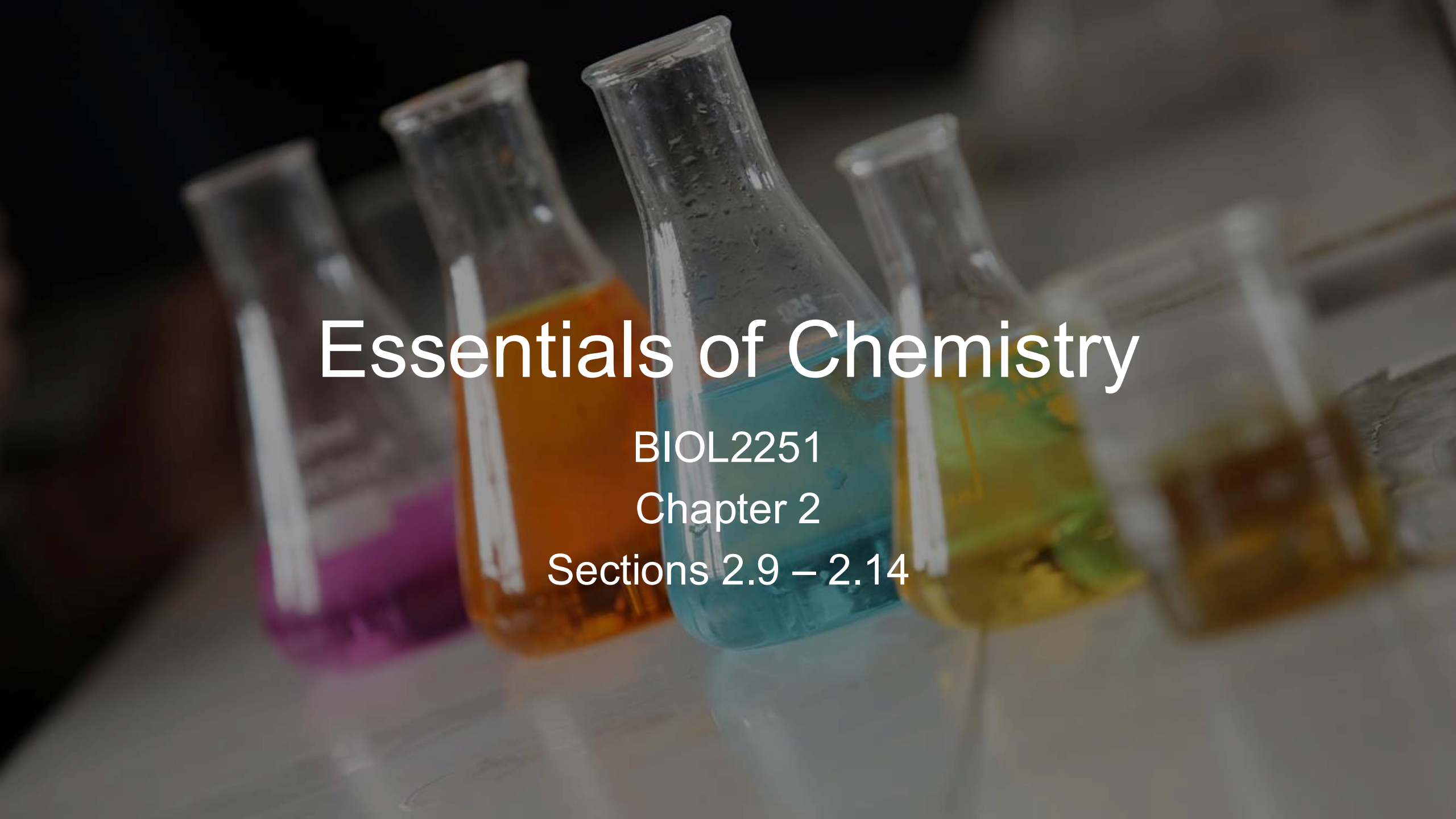


The background of the slide features four Erlenmeyer flasks arranged in a row, slightly overlapping. From left to right, they contain liquids of different colors: purple, orange, blue, and yellow. The flasks are made of clear glass and have white labels. The background is dark and out of focus, emphasizing the colorful liquids.

Essentials of Chemistry

BIOL2251

Chapter 2

Sections 2.9 – 2.14



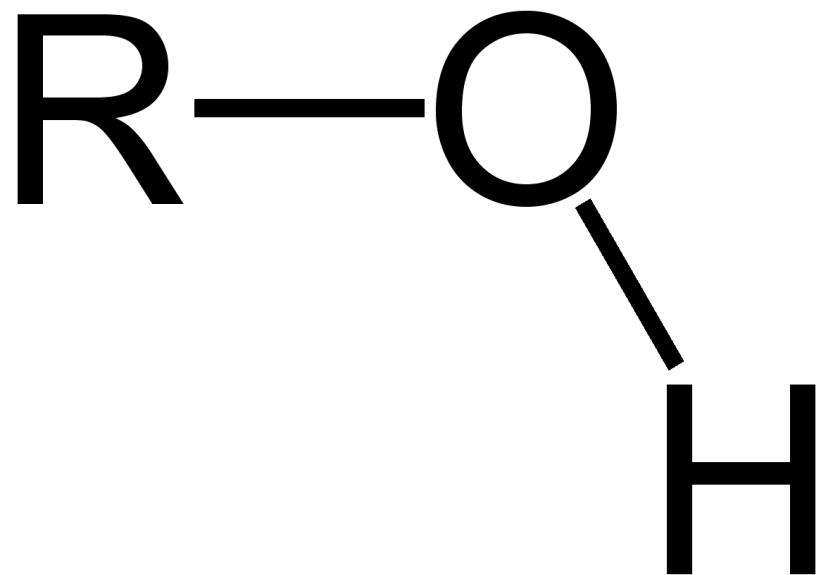
Functional Group Gallery

Functional Group Gallery

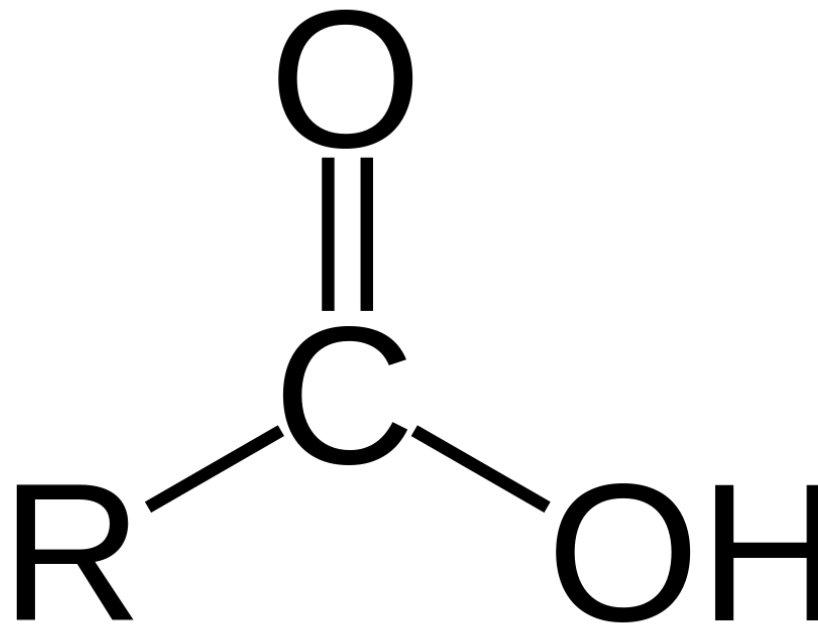
- **Instructions**

- I will project key functional groups and their chemical structure
 - **At each station, write down (1) the functional group name, (2) one property it confers, and (3) one example of a macromolecule where it can be found**
 - 2 minutes per functional group, then move through the gallery
- **Blooms taxonomy = Remember, Understand**

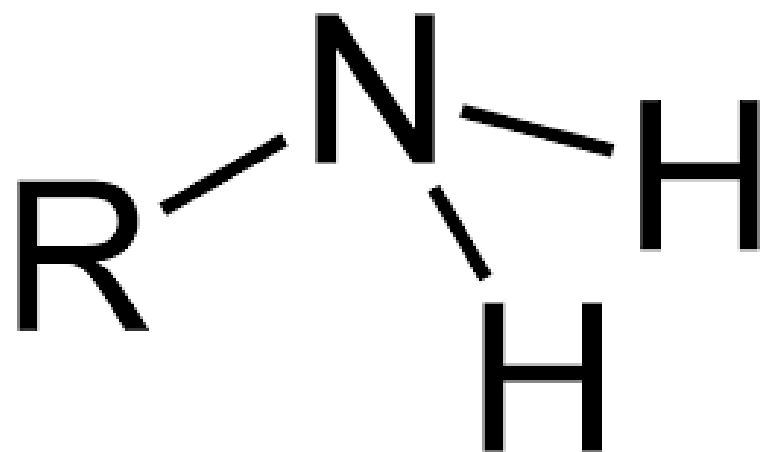
Group 1



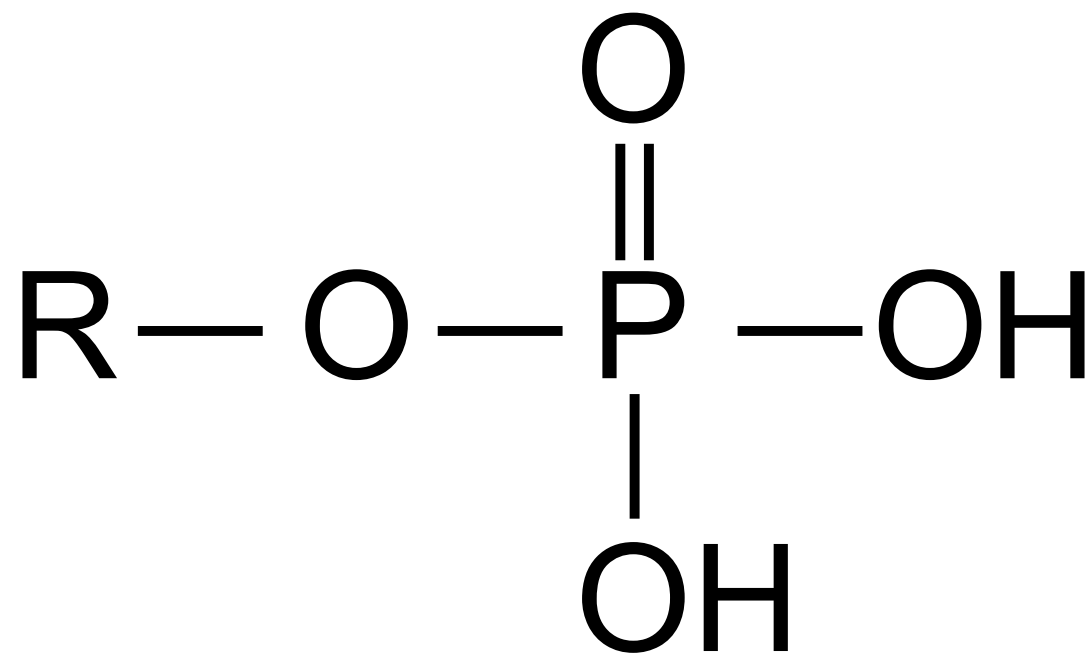
Group 2



Group 3

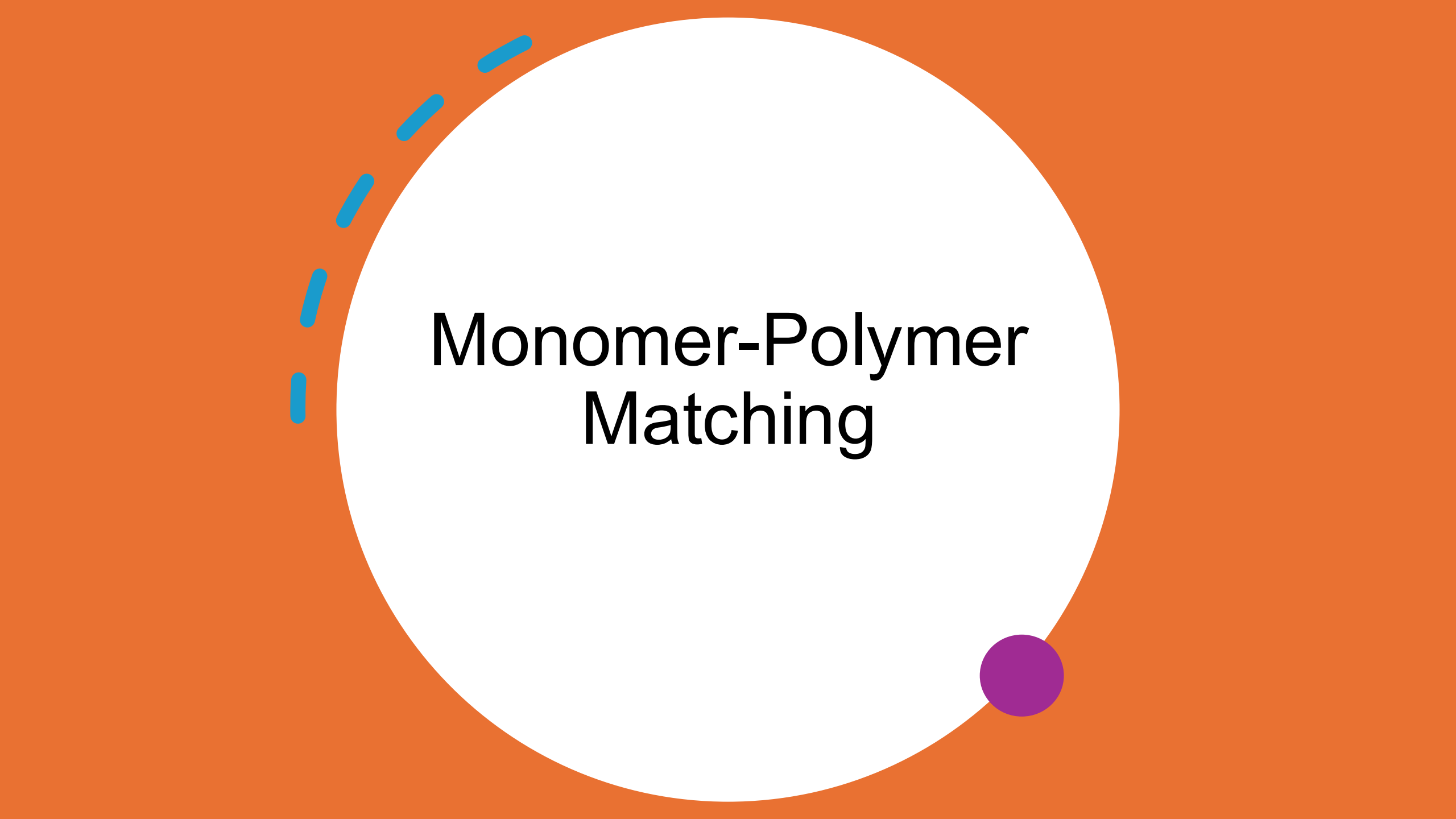


Group 4



Functional Group Gallery

- **Group 1 = Hydroxyl (-OH):** Polar, hydrophilic; found in alcohols, sugars (glucose)
- **Group 2 = Carboxyl (-COOH):** Acidic, negatively charged; found in amino acids, fatty acids
- **Group 3 = Amino (-NH₂):** Basic, positively charged; found in amino acids, nucleotides
- **Group 4 = Phosphate (-PO₄):** Negatively charged, acidic; found in ATP, nucleotides, phospholipids



Monomer-Polymer Matching

Monomer-Polymer Matching

- **Instructions**

- I will provide a list of monomers and polymers
- On your own, **take 2 minutes to match monomers and polymers and name the macromolecule class**
- We will share as a class
 - Volunteers or cold calling

- **Blooms taxonomy = Remember, Understand, Apply**

Monomer-Polymer Matching

Monomers

- Monosaccharides
- Amino acids
- Nucleotides
- Glycerol + fatty acids

Polymers

- Proteins
- Nucleic Acids
- Triglycerides
- Polysaccharides

Monomer-Polymer Matching

- **Monosaccharides** → Polysaccharides (Carbohydrates)
- **Amino acids** → Proteins (Proteins)
- **Nucleotides** → Nucleic acids (Nucleic acids)
- **Glycerol + Fatty acids** → Triglycerides (Lipids)
 - Note: Lipids don't form true polymers but are built from these components



Dehydration Synthesis vs. Hydrolysis Dramatization

Dehydration Synthesis vs. Hydrolysis Dramatization

- **Instructions**

- **We will act out the processes of dehydration synthesis and hydrolysis**
 - **I need 5 volunteers**
 - After we demonstrate the process, you will draw and label both reactions
 - Pair-share to check diagrams
- **Blooms taxonomy = Understand, Apply**

Dehydration Synthesis vs. Hydrolysis

- **Dehydration Synthesis**

- $\text{Monomer-OH} + \text{H-Monomer} \rightarrow \text{Monomer-Monomer} + \text{H}_2\text{O}$
- Builds polymers, releases water

- **Hydrolysis:**

- $\text{Monomer-Monomer} + \text{H}_2\text{O} \rightarrow \text{Monomer-OH} + \text{H-Monomer}$
- Breaks down polymers, uses water

The background features several abstract geometric elements: a large orange circle on the right, a purple circle in the upper left, a blue circle in the top right, a green L-shaped line, a blue dashed line, and a green square outline on the left.

Saturated vs. Unsaturated Lipids: Drawing Challenge

Saturated vs. Unsaturated Lipids: Drawing Challenge

- **Instructions**

- On your own, **draw a basic illustration of a saturated and an unsaturated fat**
 - Label key features: single vs. double bonds, straight vs. kinked shape
- Then **write 2-3 sentences explaining how the structure affects function**
- Compare your work with a partner
- **Blooms taxonomy = Understand, Apply**

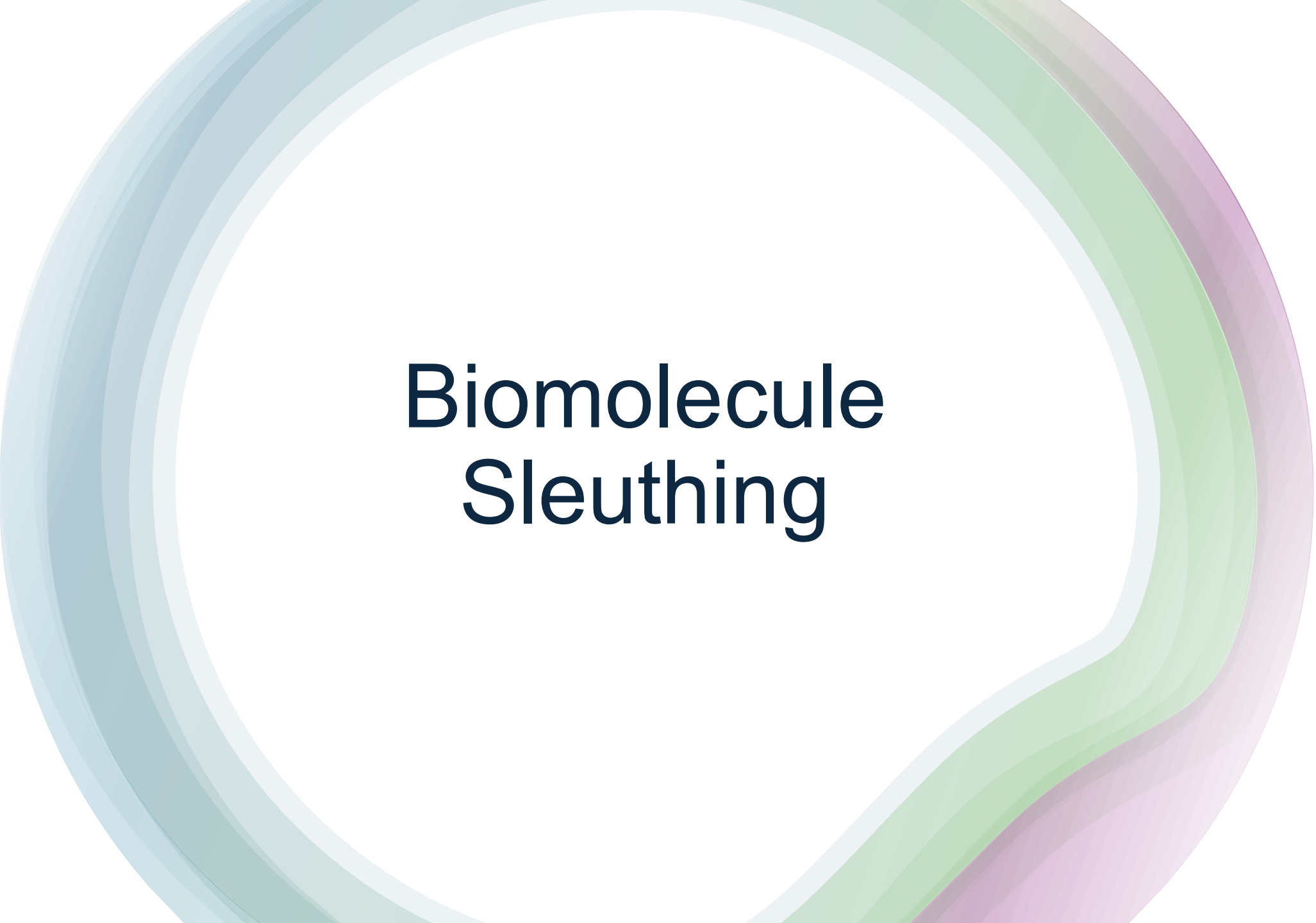
Saturated vs. Unsaturated Lipids: Drawing Challenge

Saturated

- All single bonds between carbons (C-C)
- Straight, linear chain
- Molecules pack tightly together
- Solid at room temperature (e.g., butter)
- Maximum hydrogen atoms attached

Unsaturated

- One or more double bonds between carbons (C=C)
- Kink/bend in chain at double bond site
- Molecules cannot pack tightly
- Liquid at room temperature (e.g., olive oil)
- Fewer hydrogen atoms



Biomolecule Sleuthing

Biomolecule Sleuthing

- **Instructions**

- Form 12 groups
- **Each group will receive a description of a mystery macromolecule**
 - Clues about structure, monomers, functional groups, and function
- **Identify the macromolecule as a carbohydrate, lipid, or protein**
 - **Justify your answer using evidence from the clues**
- Write your conclusion on the paper and turn it in
- **Blooms taxonomy = Understand, Analyze**

Biomolecule Sleuthing

Mystery 1

Carbohydrate (Glycogen)

- **Clues:** Made of repeating simple sugars, stores energy in animals, contains many hydroxyl groups, formula approximates $(CH_2O)_n$
- **Evidence:** Monomer = monosaccharides; functional group = hydroxyl; role = energy storage

Mystery 2

Protein (Enzyme)

- **Clues:** Made of 20 different building blocks linked by peptide bonds, contains amino and carboxyl groups, speeds up chemical reactions, shape is critical to function
- **Evidence:** Monomer = amino acids; functional groups = amino, carboxyl; role = catalysis

Biomolecule Sleuthing

Mystery 3

Lipid (Phospholipid)

- **Clues:** Has a glycerol backbone, two fatty acid tails, one phosphate group, forms cell membranes, has hydrophobic and hydrophilic regions
- **Evidence:** Not a true polymer; contains phosphate group; role = membrane structure

Mystery 4

Carbohydrate (Cellulose)

- **Clues:** Made of glucose monomers, structural support in plants, humans cannot digest it, lots of OH groups
- **Evidence:** Monomer = monosaccharide; functional group = hydroxyl; role = structure



Protein Shape- Function Case Studies

Protein Shape-Function Case Studies

- **Instructions**
 - Form 12 groups
 - **I will give you a short case study about a protein losing its function**
 - As a group, you will **analyze what happened to the shape and why function was lost**
 - Respond to the specific questions for your mini-case
 - Turn in your responses for whole class review
 - **Exit ticket!**
 - Write your first and last name on the sheet
- **Blooms taxonomy = Analyze, Evaluate**

Protein Shape-Function Case Studies

1. Shape = Function

- A protein must have the correct 3D shape to work
- Change the shape → lose the function

2. Precision Matters

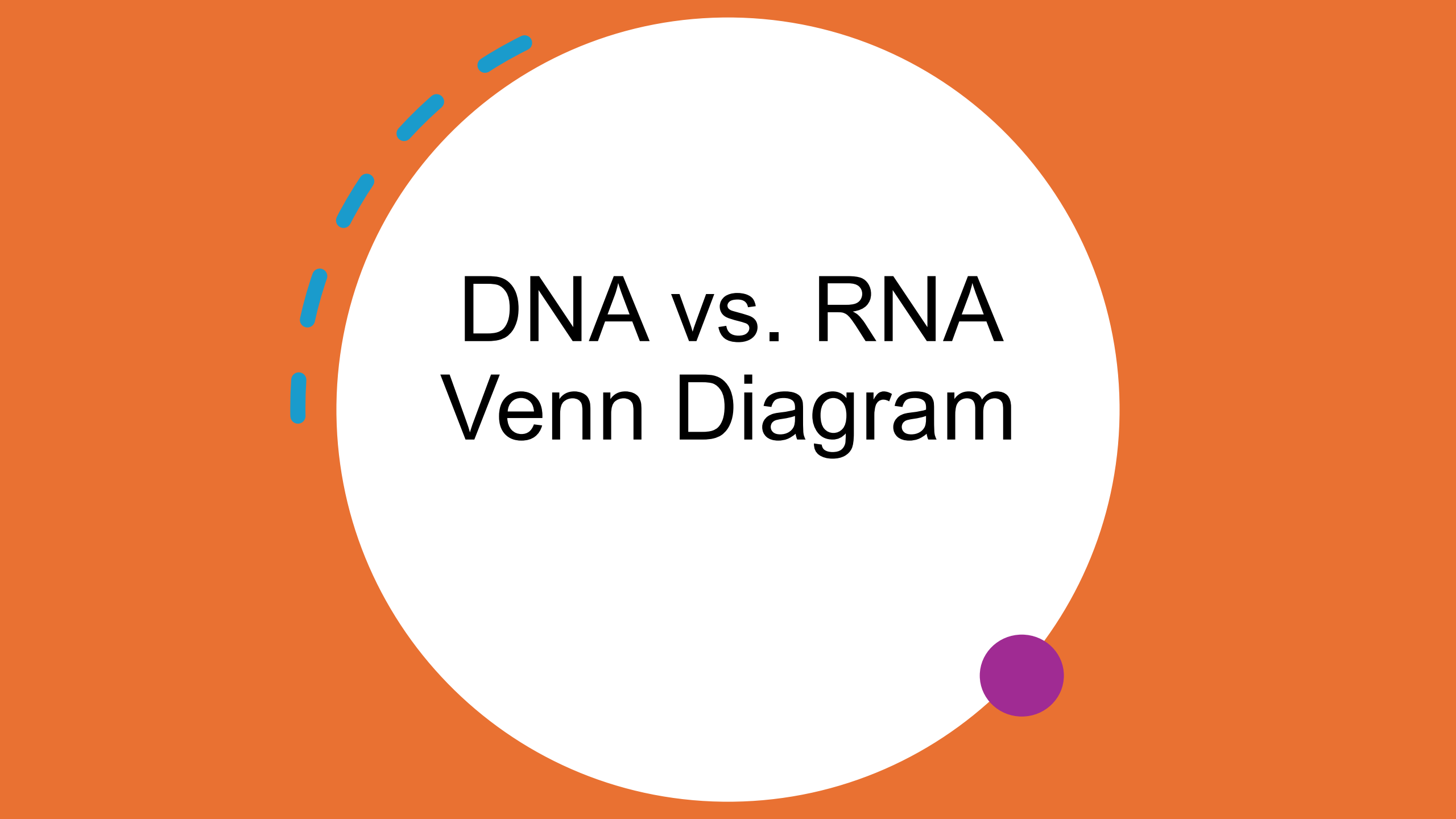
- Even small changes in shape can destroy function
- Proteins work through precise shape matching (lock and key)

3. Proteins Are Fragile

- Heat, chemicals, and pH changes can destroy protein shape
- Once unfolded (denatured), proteins usually can't refold

4. Multiple Bonds Maintain Shape

- Different types of bonds work together to hold proteins in their functional shape
- Disrupting these bonds causes proteins to lose their shape and function



DNA vs. RNA Venn Diagram

DNA vs. RNA Venn Diagram

- **Instructions**

- Work in pairs or small groups
- **Create a Venn diagram comparing DNA and RNA**
- I will set a 6-minute timer
- **List at least 4 items in each category**
 - DNA only, RNA only, Both
- Submit your work for whole class review
- **Blooms taxonomy = Understand, Analyze**

DNA vs. RNA Venn Diagram

DNA Only

- Double-stranded (double helix)
- Contains deoxyribose sugar
- Contains thymine (T)
- Stores genetic information long-term
- Found primarily in nucleus
- More stable molecule

RNA Only

- Single-stranded
- Contains ribose sugar
- Contains uracil (U)
- Transfers genetic information/makes proteins
- Found in nucleus and cytoplasm
- Less stable, temporary
- Multiple types (mRNA, tRNA, rRNA)

DNA vs. RNA Venn Diagram

Both

- Nucleic acid/polymer of nucleotides
- Contains A, G, C bases
- Contains phosphate groups
- Composed of sugar-phosphate backbone
- Made through dehydration synthesis
- Contains genetic information

ATP Energy Banking



ATP Energy Banking

- **Instructions**

- Work in small groups (3-4 people)
- **Write an analogy comparing ATP to a rechargeable battery of banking system**
- Make sure to explain: **(1) how energy gets deposited, (2) what the stored form is, and (3) how energy is withdrawn**
- Selected groups or volunteers will share
- **Blooms taxonomy = Understand, Apply**

ATP Energy Banking

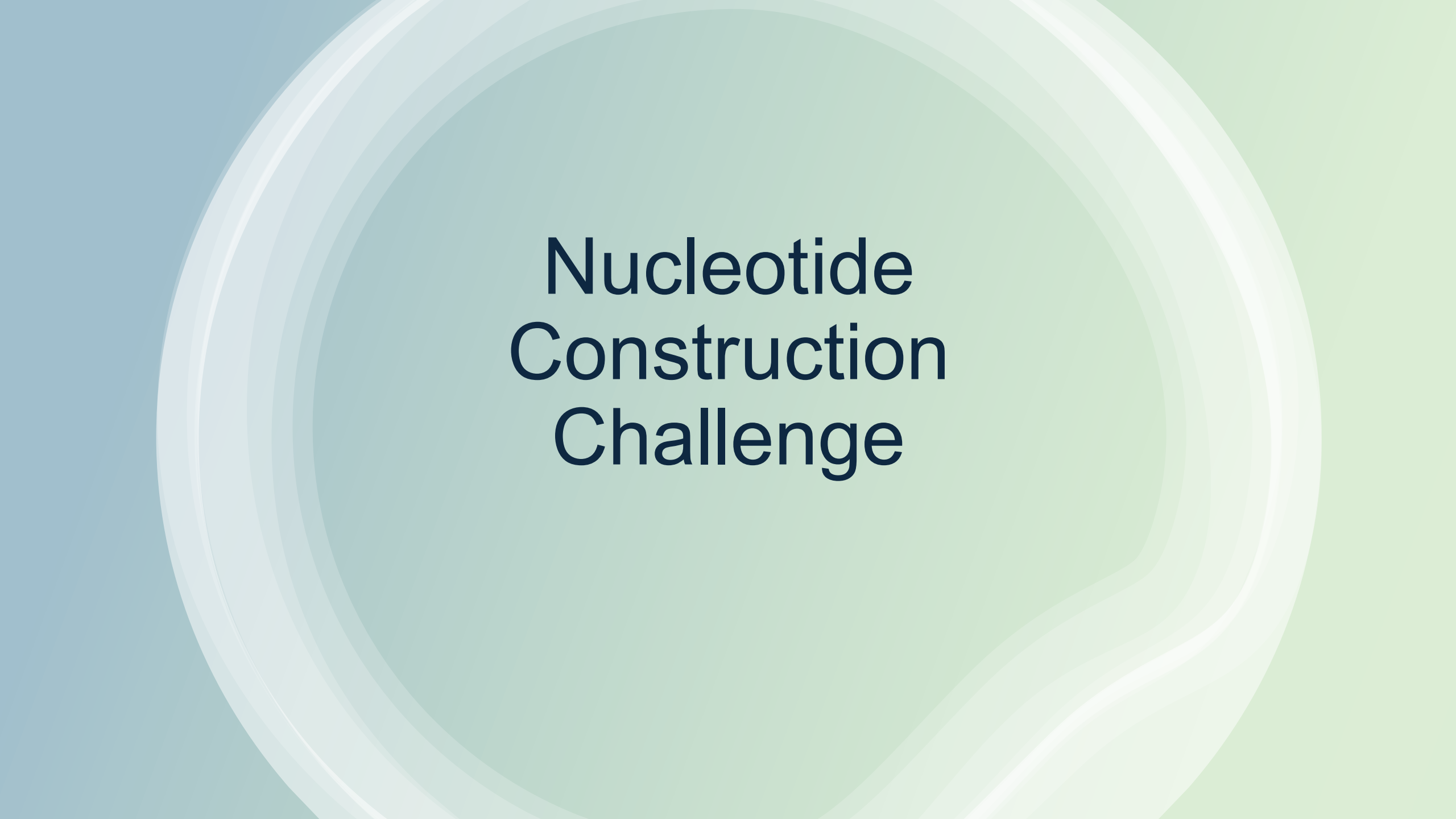
Banking Analogy:

- **Making deposit (energy storage):** $\text{ADP} + \text{P} + \text{energy (from food)} \rightarrow \text{ATP}$. Like depositing money (energy from glucose) into a bank account (ATP).
- **Stored form:** ATP molecule with three phosphate groups bonded together. The bond between the 2nd and 3rd phosphate is high-energy. Like money sitting in your account.
- **Making withdrawal (energy release):** $\text{ATP} \rightarrow \text{ADP} + \text{P} + \text{energy released}$. The terminal phosphate bond breaks (hydrolysis), releasing energy for cellular work. Like withdrawing money to pay for something.

ATP Energy Banking

Key Points:

- ATP is the "currency" of cellular energy
- The process is reversible – ATP can be regenerated from ADP
- Energy is stored in chemical bonds, specifically the phosphate bonds
- Hydrolysis breaks bonds and releases energy
- Cells constantly cycle between ATP and ADP



Nucleotide Construction Challenge

Nucleotide Construction Challenge

- **Instructions**

- Work in small groups (12 total, using whiteboards)
- **Draw and label three components of a nucleotide**
- **Modify your drawing to show (1) a DNA vs. RNA nucleotide and (2) how ATP is a 'special' nucleotide**
- All groups will hold up completed drawings/diagrams for a check-in
- **Blooms taxonomy = Remember, Understand, Apply**

Nucleotide Construction Challenge

Characteristic	Basic Nucleotide	DNA	RNA	ATP
Sugar				
Phosphate Group(s)				
Base(s)				

Nucleotide Construction Challenge

Characteristic	Basic Nucleotide	DNA	RNA	ATP
Sugar	Pentose	Deoxyribose	Ribose	Ribose
Phosphate Group(s)	1+	1	1	3
Base(s)	Nitrogenous base	A, T, C, G	A, U, C, G	A (Adenine)



Macromolecule Classification Challenge

Macromolecule Classification Challenge

- **Instructions**

- Work in pairs or small groups
- **Create a Venn diagram comparing DNA and RNA**
- I will set a 6-minute timer
- **List at least 4 items in each category**
 - DNA only, RNA only, Both
- Submit your work for whole class review
- **Blooms taxonomy = Understand, Analyze**

Biomolecule Example	Class	Monomer(s)	Key Functional Groups	Primary Role(s)
Glycogen	Carbohydrate			
Hemoglobin	Protein			
Testosterone (Steroid)	Lipid			
DNA	Nucleic acid			
Insulin	Protein			Hormone (glucose regulation)
Cholesterol	Lipid (steroid)			Membrane structure, hormone precursor
RNA	Nucleic acid			Protein synthesis
Triglyceride	Lipid			
Collagen	Protein			Structural support (connective tissue)
ATP	Nucleic acid	Adenosine nucleotide		

Biomolecule Example	Class	Monomer(s)	Key Functional Groups	Primary Role(s)
Glycogen	Carbohydrate	Glucose	Hydroxyl (-OH)	Energy storage (animals)
Hemoglobin	Protein	Amino acids	Amino (-NH ₂), Carboxyl (-COOH)	Oxygen transport
Testosterone (Steroid)	Lipid	N/A (not a polymer)	Hydroxyl (-OH)	Hormone signaling
DNA	Nucleic acid	Nucleotides	Phosphate (-PO ₄), Amino	Genetic information storage
Insulin	Protein	Amino acids	Amino, Carboxyl	Hormone (glucose regulation)
Cholesterol	Lipid (steroid)	N/A	Hydroxyl (-OH)	Membrane structure, hormone precursor
RNA	Nucleic acid	Nucleotides	Phosphate (-PO ₄), Hydroxyl, Amino	Protein synthesis
Triglyceride	Lipid	Glycerol + 3 fatty acids	Carboxyl (-COOH)	Energy storage, insulation
Collagen	Protein	Amino acids	Amino, Carboxyl	Structural support (connective tissue)
ATP	Nucleic acid	Adenosine nucleotide	Phosphate (-PO ₄) x3, Amino	Energy transfer